LEVELHULL · DESIGN DECISION RECORD

LevelHull wider foam modules for R12 and R13

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DOCUMENT CONTROL

Document ID	LVH-DDR-003	Version	0.1
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/decisions/0003-wider-modules-r12-r13.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Decision 10A recorded and carried into the model, calculations, bill of materials, drawings and build plan

0003: Wider foam modules for R12 and R13

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish on 2026-10-03, choosing option A on every requirement decision put to him: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A". This record carries decision 10A.

Context

At TRL 3 (LVH-CAL-001 v0.1) R12 and R13 were met on paper, but with little margin. With the largest module lost and 5 % water uptake the lowest point of the gunwale was only 9 mm above the water (R12). With all four crew holding one grab line the heel was 4.0 deg, inside the 10 deg of R13, but the low-side gunwale was only 31 mm above the water. Both had been handled as operating rules rather than by the design.

Options considered

- **A. Enlarge the foam modules about 10 % (chosen).** More lift and more waterplane along both sides for a small rise in foam, cover and mass.
- **B.** Keep the modules and keep the operating rules (crews spread out, no fishing with a module missing).

Decision

Option A. The modules are widened from 200 to 220 mm (level width), keeping the 200 mm height of four 50 mm layers, the lengths, the positions high under the gunwale and every fixing. That is 10 % more foam: 0.42 m3 of module in all, against 0.38 m3.

Width rather than height or length was chosen because:

- A fifth layer would add 25 %, not 10 %, and would bring the module bottom down towards the floor, where it adds less roll stiffness.
- The lengths are already set by the thwarts and frames.
- Four 220 mm strips still come from the 1,000 mm width of one foam sheet, so the six sheets per kit stay the same. A new model check confirms it.
- The battens, chocks and screws are unchanged. Each strap goes 40 mm further round the module, so straps are cut 1.25 m long instead of 1.2 m (20 m of the 25 m bought), and each cover panel is cut 1,040 mm wide instead of 1,000 mm.

Results (LVH-CAL-001 v0.2)

Requirement	Target	Before	After
R12	Still floats with the largest module lost and 5 % uptake	9 mm lowest freeboard, 1.5 deg heel	25 mm lowest freeboard, 1.5 deg heel
R13	Heel within 10 deg, all four crew on one grab line	4.0 deg; low side 31 mm	3.9 deg; low side 44 mm
R1, R11	Afloat with design load; reserve at least 1.5	2.17	2.45
R2	At least 50 mm freeboard	80 mm	92 mm
R5	10 % or less of the inside volume	7.6 %	8.4 %
R7	Fixings at least 2 times their share	Lowest 5.5	Lowest 5.0
R10	Value-engineering target USD 1,000 for two canoes	USD 1,350	USD 1,360

Kit mass rises from 67.4 to 68.9 kg (foam 12.1 kg, covers 9.3 kg). The clear width between the module tops falls from 963 to 923 mm, still above the 900 mm the model checks. The model passes 127 of 127 constructability checks.

Consequences

- Value-engineering target: USD 1,000. Estimated cost of the constructable design: USD 1,360 (USD 360 over the target). The only added cost is 1 m2 more tarpaulin a kit (USD 5); the foam strips come from the same six sheets, so no spare strip is left.
- With all four crew on one side the low gunwale is still under 50 mm from the water, so the operating rule that crews spread along both grab lines stays.
- The comparison with the same foam on the floor changes: the larger volume now just reaches the surface on the floor, but it heels 10.6 deg with the crew on one side and has no roll stiffness left with 5 kg more load. The case for fitting the modules high (LVH-DDR-001, D1) stands.
- The cores, covers, general arrangement (Rev P3), making sketches of the cores and covers, the module joint pictures, the step pictures and the concept media are redrawn from the model.

> **Safety:** Wider modules give more margin when the canoe is swamped; they do not make it safe. LevelHull is never a substitute for life jackets, and the swamp test rules in the build plan are unchanged.